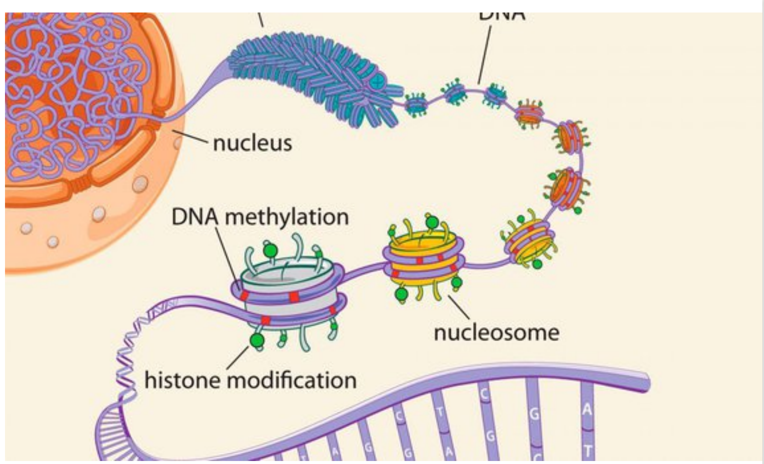
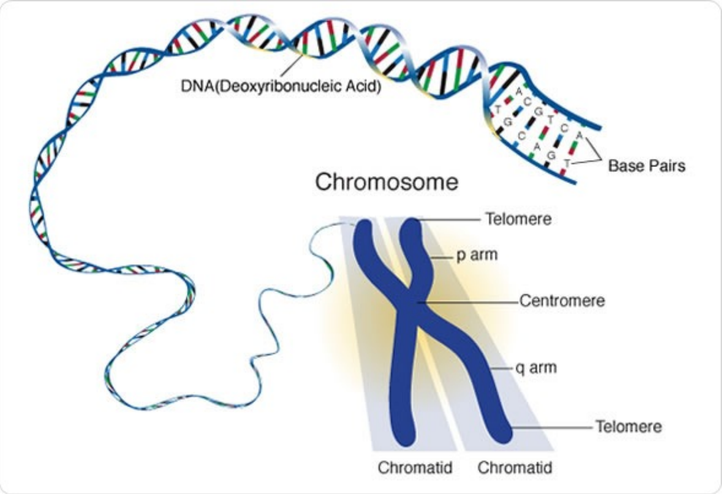
Introduction to Bioengineering Theory (BBL1020; 3-0-0)

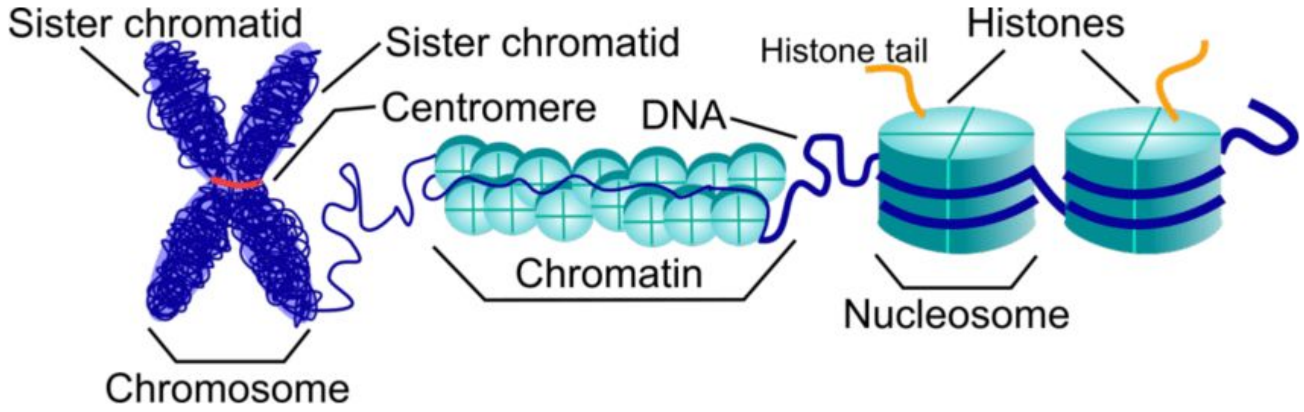
Lecture-13

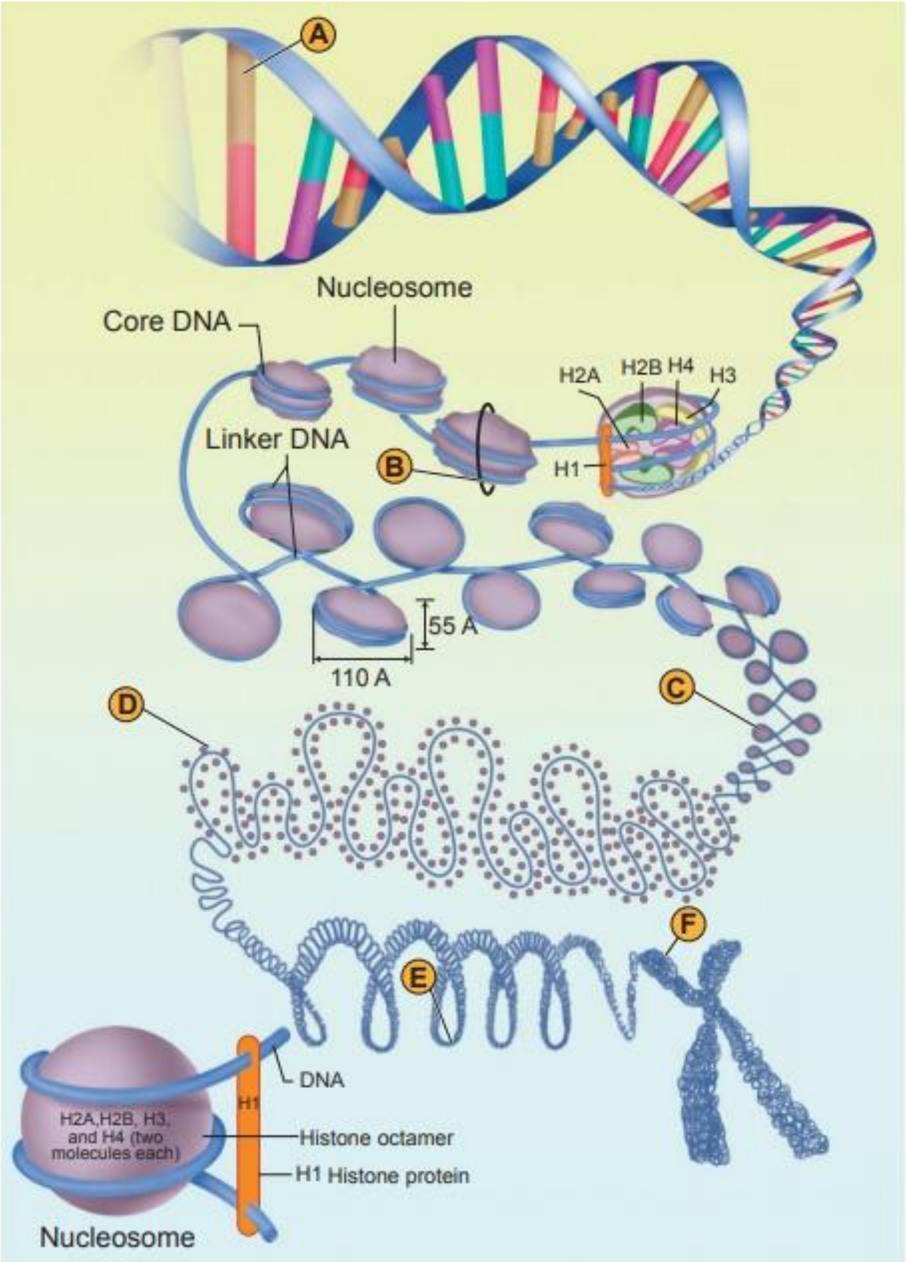
Please keep your phone(s) on silent mode

Packaging of genome

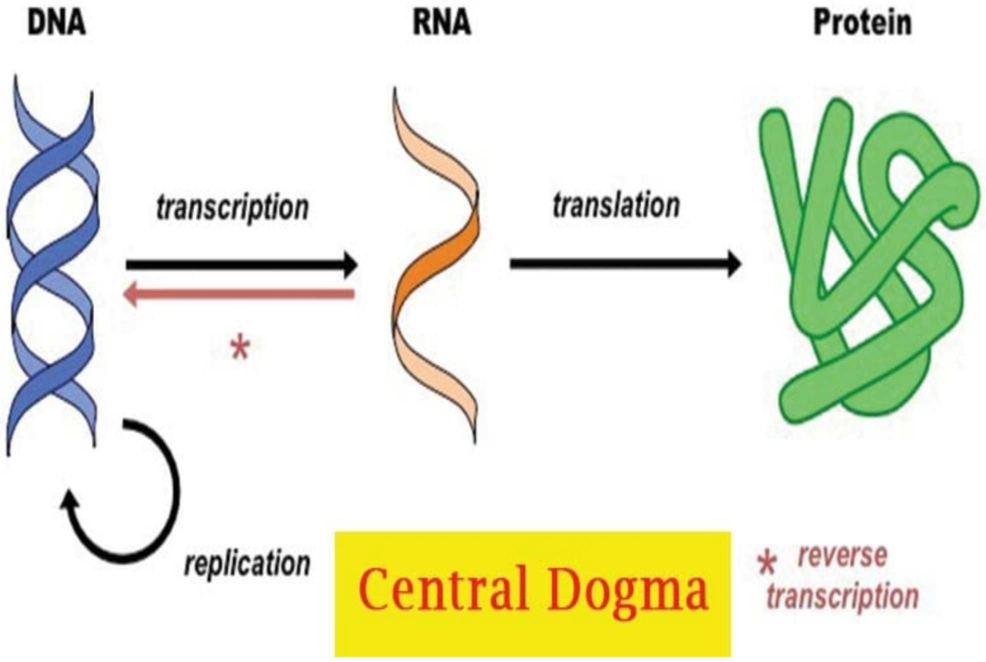








Condensation of DNA - A - DNA, B-Nucleosomes and Histones, C- Chromatin fiber, D- Coiled chromatin fiber, E- Coiled coil, F- metaphase chromatid



Importance of the Central Dogma

- Framework for Understanding Genetic Information Flow from DNA to RNA to proteins - development, functioning, and reproduction of all living organisms.

- Basis for Molecular Genetics: to understand how genes are expressed and regulated- development of genetic engineering, gene therapy, and the CRISPR-Cas9 gene-editing technology.

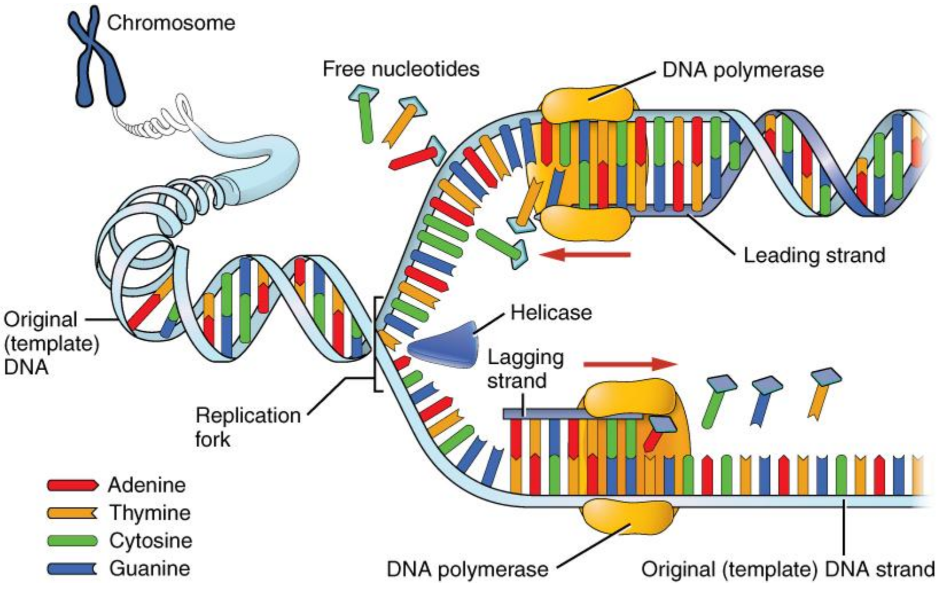
- Insights into Evolution: The conservation of these mechanisms across diverse forms of life highlights their importance in the evolution of biological systems.

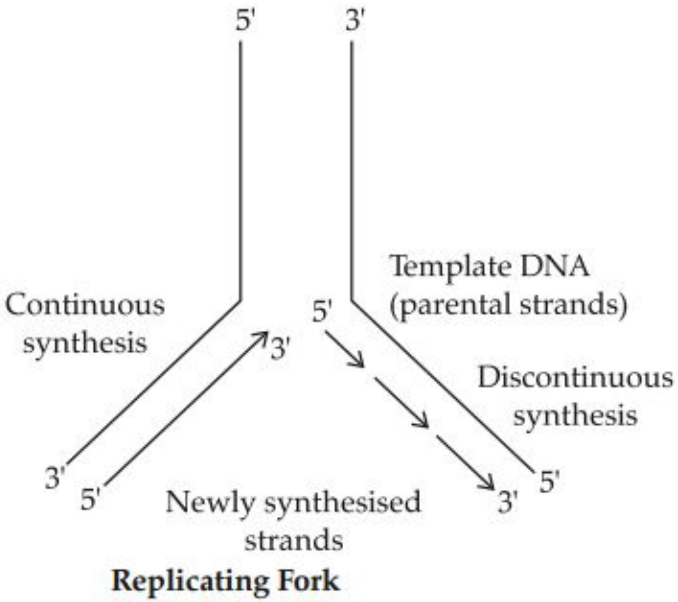
- Medical Research and Diagnostics: identifying the genetic basis of diseases, better diagnostic tools, treatments, and the development of personalized medicine.

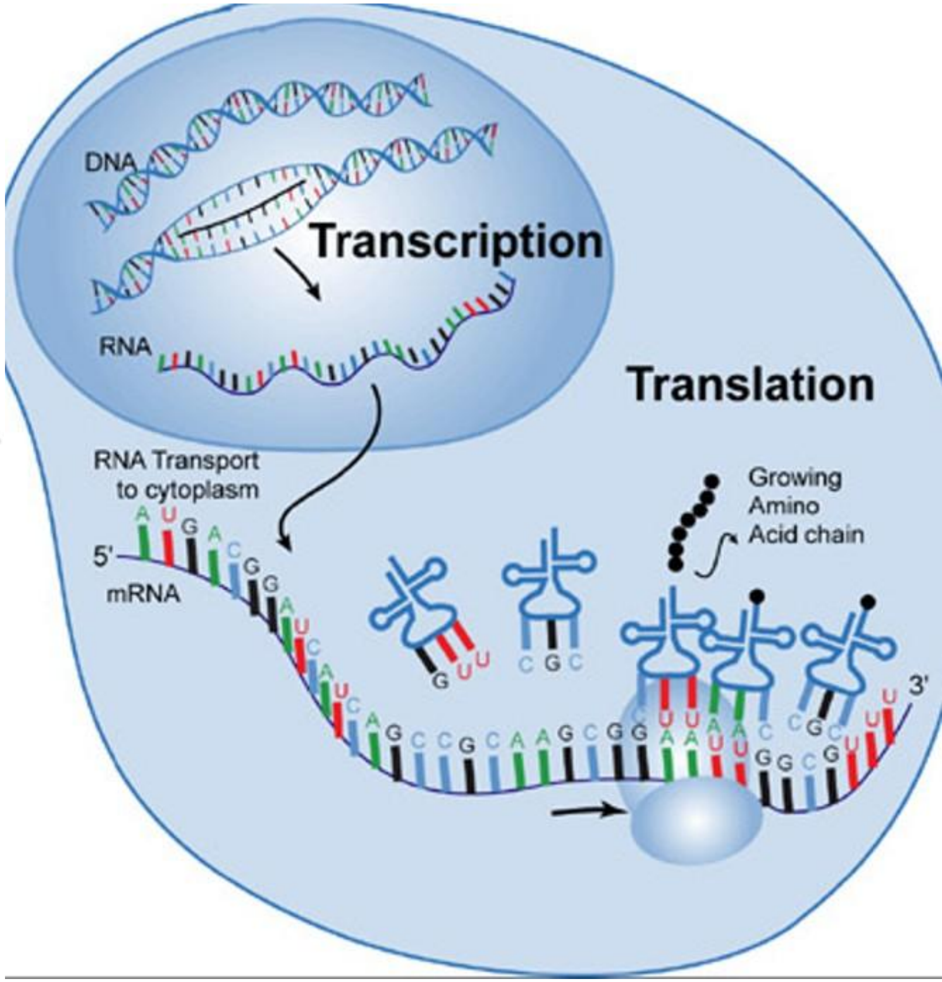
- Biotechnology and Synthetic Biology: design of synthetic genes to produce new proteins with desired functions- medicine, agriculture, and industry eg insulin, growth hormones, and biofuels.

- Understanding RNA's Role - new fields of research and therapeutic strategies.

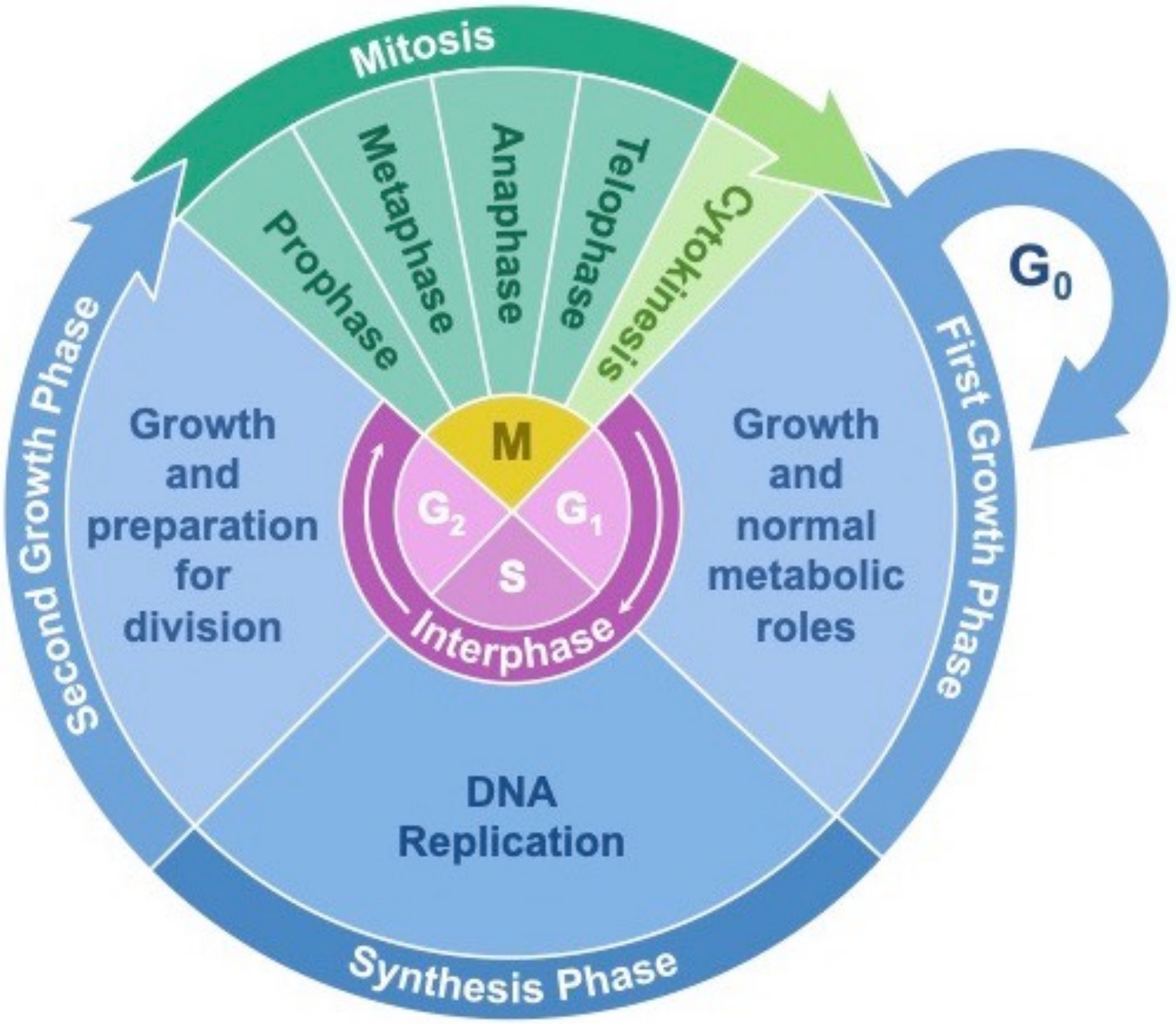
Central Dogma of Biology







Cell Division



STAGES OF THE CELL CYCLE

INTERPHASE:

G_1 – Growth and metabolic roles
S – Replication of DNA occurs
G_2 – Growth and more preparation

MITOSIS:

P – Chromosomes are condensed
M – Chromosomes align at cell centre
A – The duplicated DNA segregates
T – Chromosomes are decondensed

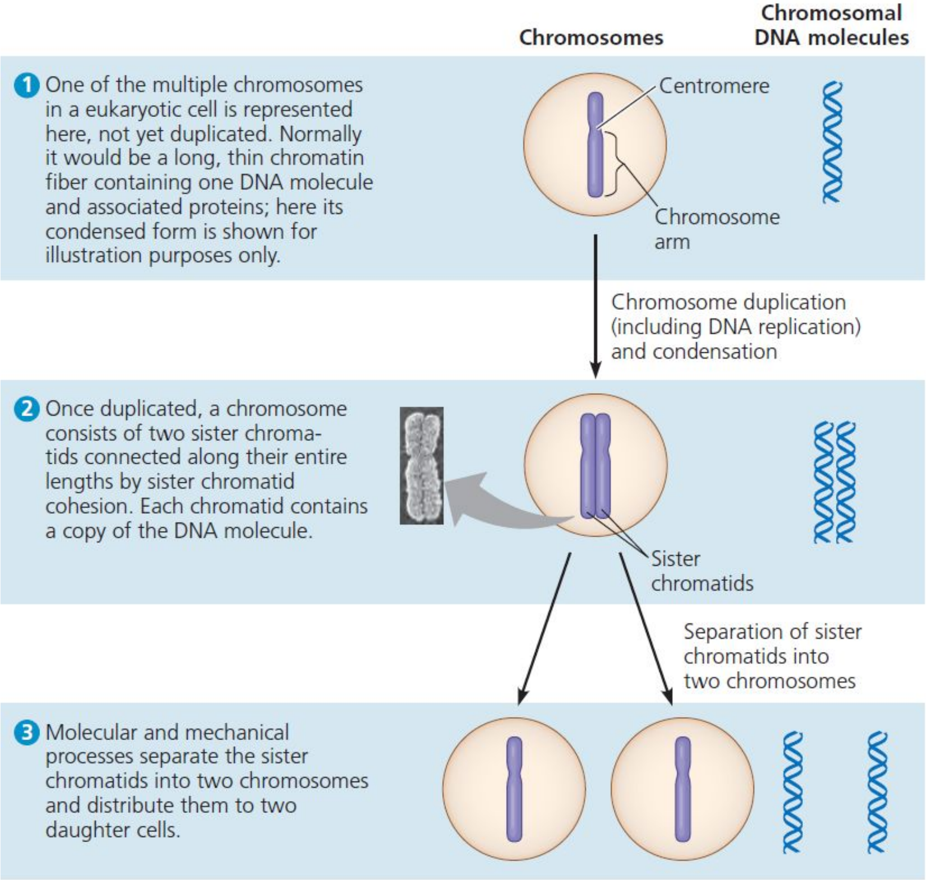
CYTOKINESIS

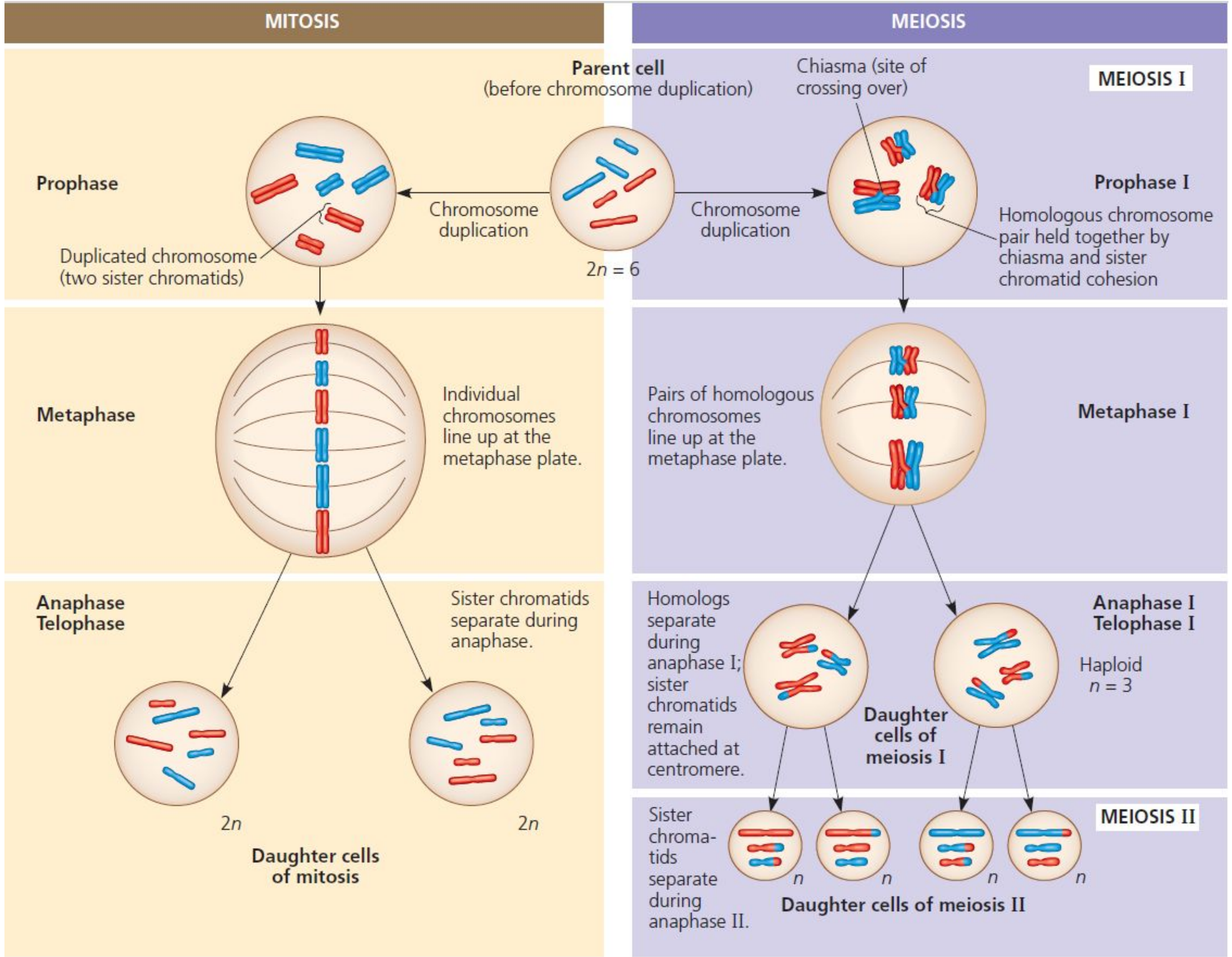
Cell splits into two daughter cells

RESTING PHASE (G_0)

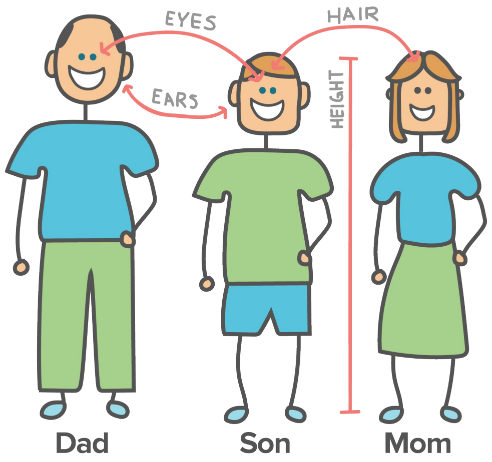
Cells may leave interphase and enter Into a non-dividing quiescent phase

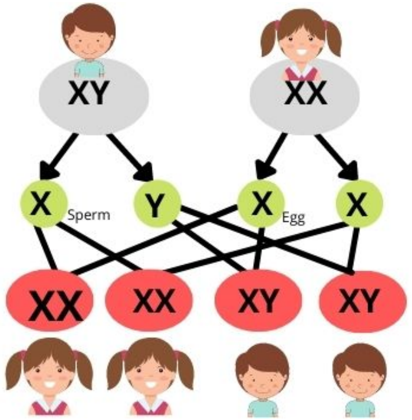
Cell Division

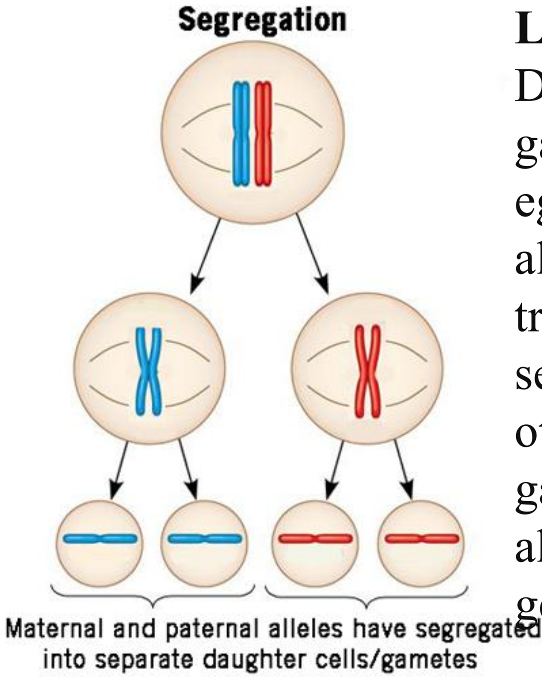




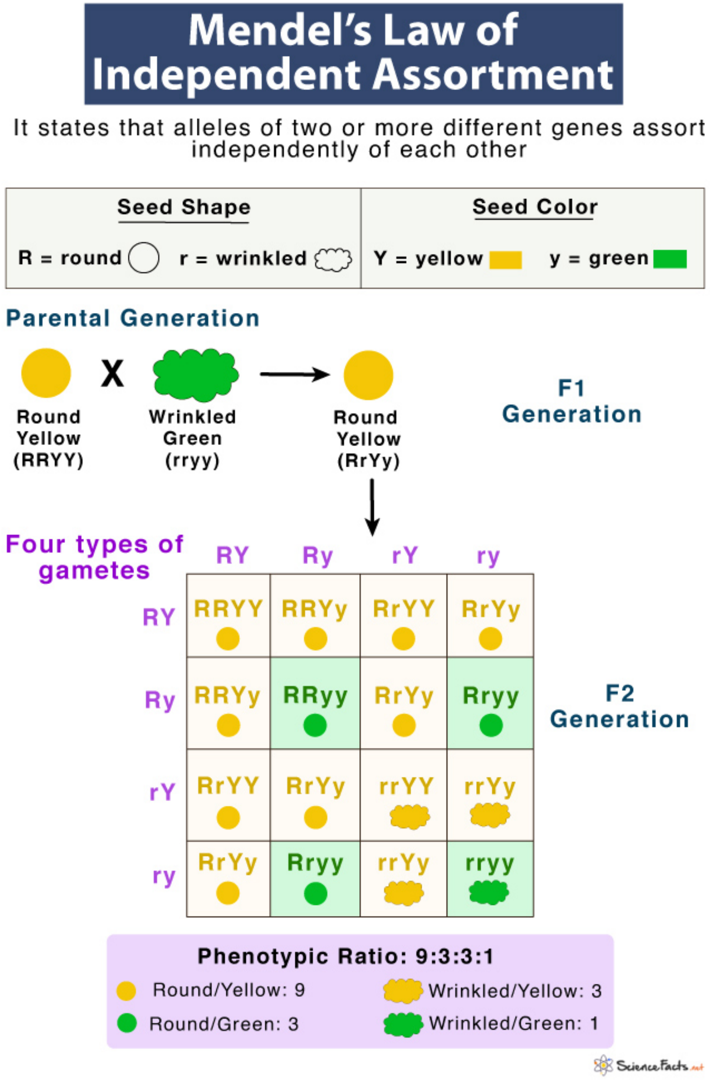
Genetic Inheritance





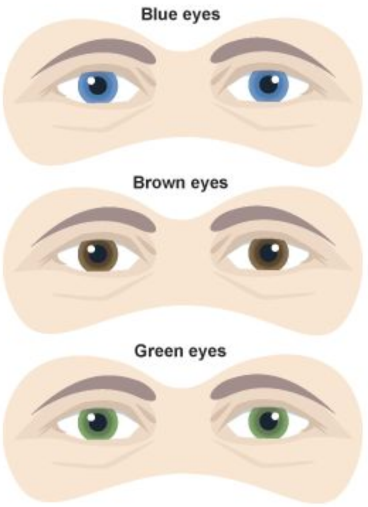


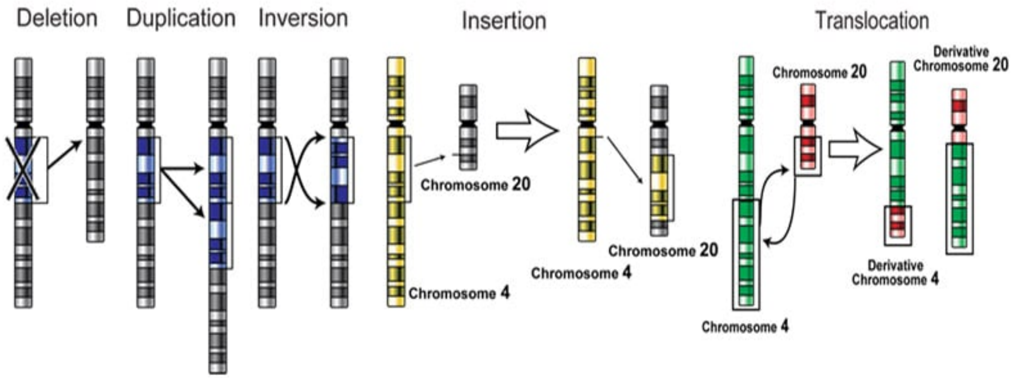
Law of segregation: During the formation of gametes (sperm and egg cells), the two alleles for a hereditary trait segregate or separate from each other. As a result, each gamete carries only one allele for a particular gene.

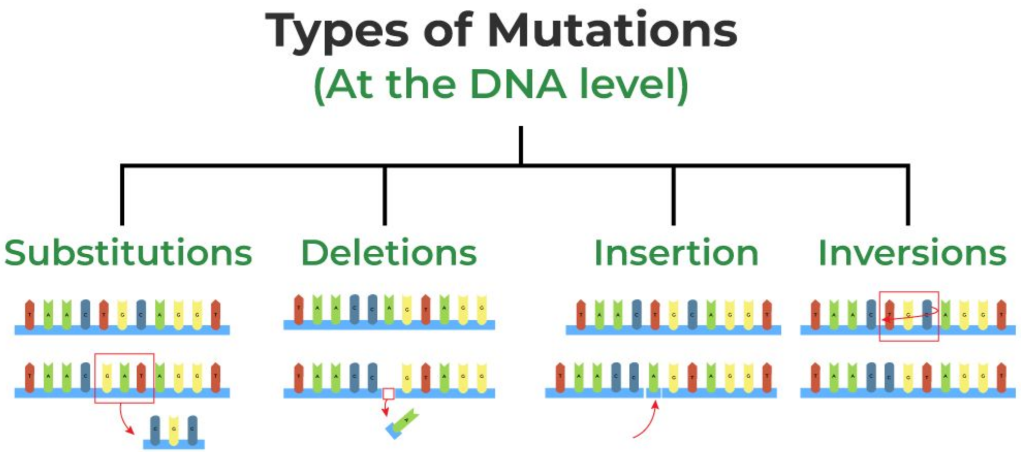


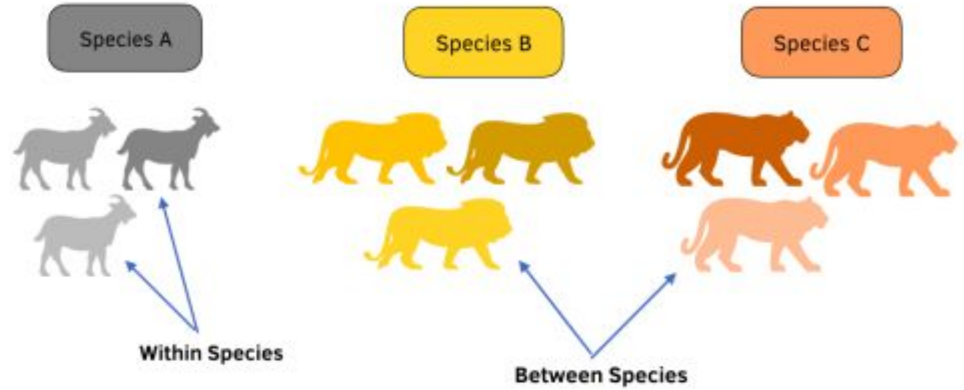
Law of independent assortment: The inheritance of an allele for one gene does not influence the inheritance of alleles for other genes.

Genetic Variation









- Mutation
- Gene flow
- Sexual reproduction
- Genetic drift
- Random mating
- Crossing over
- Environmental variance